

Model 1 – Three-Hour Flash-Flood Probability

TECHNICAL MODEL REPORT | PRAVAAH | GENERATED 30 SEPTEMBER 2026

1. Executive summary

Model 1 is a Random Forest binary classifier that maps 14 rainfall, soil, terrain, hydrology, and historical-event inputs to one probability in the range 0–1 for a flash-flood event within the next three hours. The artifact is reproducible, validated at the input boundary, and currently trained on generated synthetic data; reported metrics indicate prototype behaviour only.

2. Dataset and target definition

Rows: 6,000 | **Train/test split:** 80%/20% with stratification and `random_state = 42`.

Target: event (0 = no event, 1 = event); class counts are 538 negative and 5,462 positive. The generator uses seeded Gamma, Beta, Normal, log-normal, uniform, and Poisson distributions, then derives a latent risk score and samples labels.

Input features and meaning:

- **Rainfall accumulation:** `rain_1h`, `rain_3h`, `rain_6h`, `rain_24h`, `rain_72h`
- **Rainfall dynamics:** `max_rain_intensity`, `rainfall_rate`
- **Hydrologic state:** `soil_moisture`, `soil_moisture_change`, `flow_accumulation`, `twi`
- **Terrain/exposure:** `slope`, `distance_to_river`
- **Historical context:** `historical_flood_count`

3. Model configuration

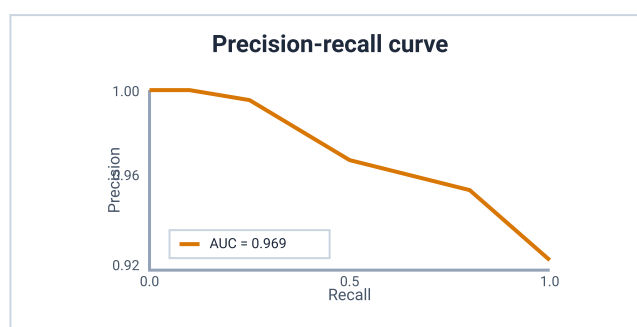
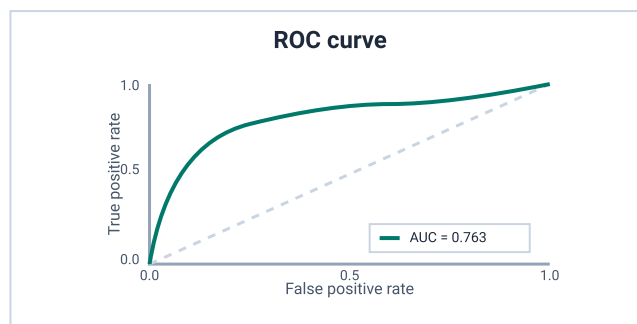
Algorithm	Estimators	Max depth
Random ForestClassifier	300 trees	12
Min samples leaf	Class weighting	Seed
3	balanced_subsample	20260926
Decision threshold	Output	
0.50 for metrics	P(event within 3 h)	

Model 1 – Validation Results

HOLDOUT EVALUATION ON THE SYNTHETIC PROTOTYPE DATASET

5. Classification metrics

Metric	Score	Percent
Accuracy	0.887	88.7%
Precision	0.919	91.9%
Recall / Sensitivity	0.960	96.0%
F1 score	0.939	93.9%
ROC-AUC	0.763	76.3%
PR-AUC	0.969	96.9%



Confusion matrix (actual vs predicted)

	No event	Event	
No event	16	92	Actual
Event	44	1048	
	No event	Event	
	Predicted		

6. Top feature importance

rain_1h		0.113
rain_6h		0.106
max_rain_intensity		0.106
rain_3h		0.083
rain_72h		0.082
soil_moisture		0.073
twi		0.072
rain_24h		0.066

7. Interpretation and limitations

Metrics are from a single stratified 80/20 holdout and a separately fitted validation forest; they are not cross-validation estimates and may be optimistic because the data and labels are synthetic. Feature importance is model-relative, not causal. Before operational use, replace generated data with verified event observations, perform spatial/temporal leakage checks, calibrate probabilities, select thresholds with emergency-management stakeholders, and validate on an external region/time period.

Model 2 — Landslide Susceptibility & Dynamic Risk

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1. Executive summary

Model 2 is a Random Forest classifier that maps 17 terrain, soil, land-cover, vegetation, and historical-rainfall inputs to a landslide susceptibility score in the range 0–1 for geographic cells in Uttarakhand. A separate dynamic risk engine combines the static susceptibility with recent rainfall and soil-moisture conditions to produce a current landslide risk score. The model is a research/prototype system; the reported metrics are internal development evaluation, not an operational warning-system score.

2. Dataset and target definition

Region/grid: Uttarakhand, India | 52,674 cells at 1 km × 1 km. **Inventory:** 5,523 records extracted; 5,510 inside the state boundary were used, 13 outside were excluded.

Target: 1 = at least one historical landslide inventory point in the cell; 0 = background cell with no inventory point (not proof that no landslide occurred). Positive cells 3,079 (5.85%); background 49,595.

Split: group-based spatial split, train 42,254 / test 10,420 cells (~80/20), random_state = 42, spatial-group overlap = 0.

Sources: ISRO/NRSC Landslide Atlas and GSI/Bhuvan (inventory); USGS SRTMGL1_003 (DEM); MERIT Hydro; OpenLandMap (soil); ESA WorldCover; Sentinel-2 NDVI; CHIRPS Daily (historical rainfall); NASA SMAP (dynamic soil moisture).

Static input features (17) and meaning:

Terrain/topography: elevation, slope, aspect_sin, aspect_cos, curvature

Hydrology/drainage: upstream_area, TWI, drainage_density, distance_to_drainage

Soil: clay_surface, organic_carbon_surface

Land/vegetation: landcover, NDVI

Historical rainfall: mean_annual_rainfall, max_daily_rainfall, mean_monsoon_rainfall, high_rain_days

Soil moisture and historical landslide count are deliberately excluded from the static model.

3. Model configuration

Algorithm	Static → processed features	Decision threshold
Random Forest Classifier	17 → 26 (after encoding)	0.40
Training cells	Spatial test cells	Output
42,254	10,420	P(landslide susceptibility)

Preprocessing: 16 numeric features median-imputed + standardized; landcover most-frequent-imputed + one-hot. Random Forest chosen after comparing Logistic Regression and XGBoost; retrained on all 42,254 training cells. Hyperparameters not listed (not in supplied config).

4. Static susceptibility vs dynamic risk



Dynamic trigger = $0.70 \times \text{rainfall} + 0.30 \times \text{soil moisture}$; rainfall = $0.60 \times 3\text{-day} + 0.40 \times 15\text{-day}$ (1-day kept as context).

Current risk = susceptibility × (1 + 0.50 × trigger), clipped to 0–1. Prototype categories: Low <0.20 | Moderate 0.20–<0.40 | High 0.40–<0.60 | Very High ≥0.60. Weights and thresholds are prototype rules, not validated emergency-warning thresholds.

Full-grid descriptive output (52,674 cells; not validation metrics): susceptibility mean 0.1857, max 0.9506, 11,265 cells ≥ 0.40; trigger mean 0.6276 (std 0.1274); current risk mean 0.2468, median 0.0730. **Category counts:** Low 31,014 | Moderate 5,711 | High 7,148 | Very High 8,801.

Model 2 — Validation Results

INTERNAL SPATIAL VALIDATION OF THE CORRECTED STATIC MODEL

5. Classification metrics

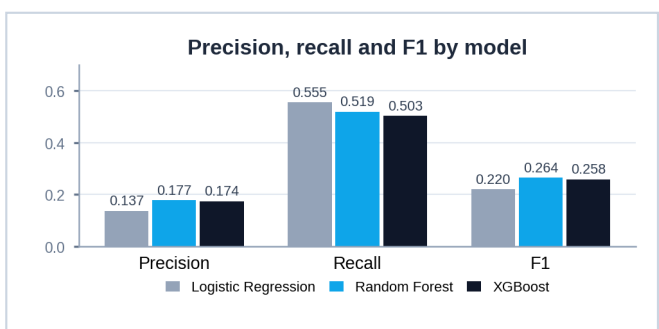
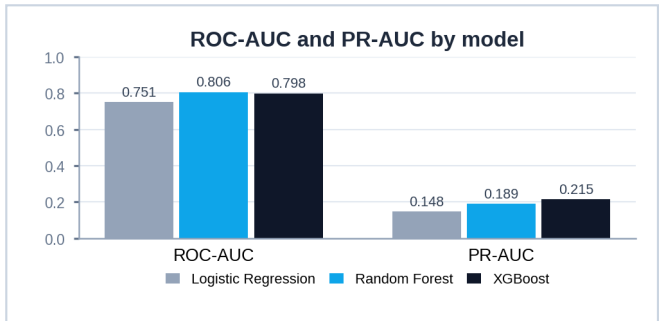
Metric	Score	Percent
PR-AUC	0.1894	18.94%
ROC-AUC	0.8064	80.64%
Precision	0.1772	17.72%
Recall / Sensitivity	0.5190	51.90%
F1 score	0.2642	26.42%
Decision threshold	0.40	—

Corrected static model. Accuracy and confusion-matrix counts are not available for this evaluation.

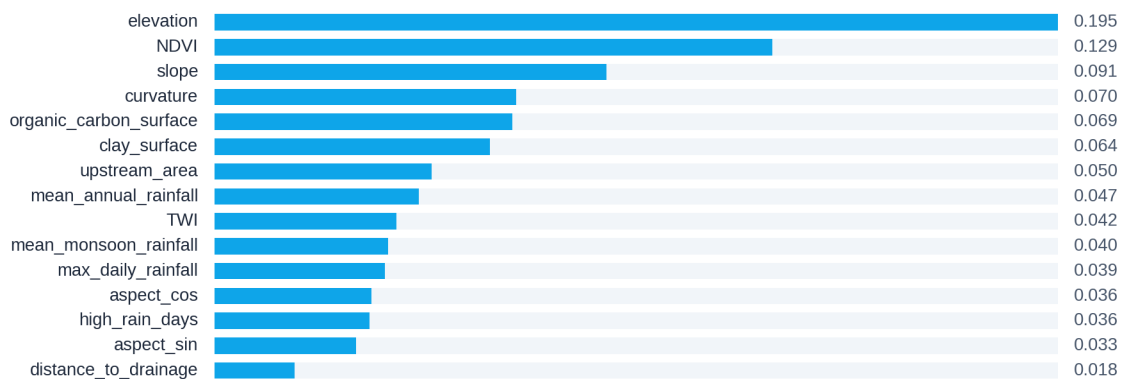
Model comparison (same internal validation)

Model	ROC	PR	Thr	Prec	Rec	F1
Logistic Reg.	0.7513	0.1483	0.60	0.1369	0.5552	0.2196
Random Forest	0.8064	0.1894	0.40	0.1772	0.5190	0.2642
XGBoost	0.7984	0.2148	0.45	0.1739	0.5027	0.2584

Random Forest was selected as the final model for the configured development objective.



6. Top feature importance



15 features shown; landcover is one-hot encoded and has no single importance value. Importance is model-relative, not causal.

7. Interpretation and limitations

- Inventory is incomplete; background cells do not prove that no landslide occurred. Susceptibility is **not** the probability of a landslide tomorrow.
- Input resolutions differ (SRTM ~30 m, MERIT Hydro ~92.8 m, soil ~250 m, CHIRPS ~5.6 km, SMAP ~9 km) and were combined on a 1 km grid; coarse CHIRPS can give block-like rainfall patterns.
- NDVI used scene-level cloud filtering without pixel-level cloud/shadow masking. Curvature is a smoothed DEM-derived proxy, not standard profile/plan curvature.
- Metrics are internal spatial-validation results, not an official SIH or field-test score. The dynamic layer's weights/thresholds are prototype rules and not operationally validated.
- Dynamic data are not guaranteed live (prototype run: CHIRPS through 2026-08-31, SMAP through 2026-09-25). Independent spatial/temporal validation is required before any operational warning use.

Colab: <https://colab.research.google.com/drive/1nuoevJdiYS8lUfWibtCJUGe3Rycoz8gW?authuser=1>

GitHub: https://github.com/Preyans-alt/SIH_Round1 — Model 2 artifacts and predictor are in models/model2 (scikit-learn 1.6.1).